Saumya Rishikesh

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Education

Manipal University Jaipur — **Jaipur, Rajasthan**
Bachelor of Technology in Computer Science — *September 2022 – July 2026*

Experience

Associate Software Engineer (AIML) — **July 2026 – Present**
Cyfuture India — *Noida, India*

- Developing a GPU-backed video-generation platform using Kubernetes (K3s), Docker, FastAPI, Redis, and ComfyUI/Wan2.2 on an NVIDIA V100; deployed separate API, worker, queue, and inference services with persistent job and video storage.
- Extending the platform with priority-aware FIFO scheduling, atomic job claims, retries, and abandoned-job recovery; roadmap covers resource allocation, Secrets, networking, monitoring, and backup/recovery. Configured GPU allocation and health probes for the deployed services.
- Designed refund and stock-transfer modules for JCI Sahyog using Java, Spring MVC, Spring Boot, and JSP; implemented multi-level approval workflows for contract refunds and payment methods.
- Contributed to CSPDCL's C#-to-Java/Spring Boot migration, building EODB and management-dashboard features and a middleware API to route client requests and retrieve location-based electricity-consumption data.

Software Engineering Intern — **January 2026 – July 2026**
DealMatrix Services Pvt Ltd — *Remote*

- Developed multilingual AI voice agents supporting **10+ Indian languages** for automated tele-calling, reducing per-call operational costs by **30–40%** versus human callers with 24/7 availability.
- Architected a streaming pipeline with Sarvam AI STT/TTS, Gemini as the primary LLM, Groq fallback, and LiveKit over WebRTC; achieved **sub-second responses** and tracked per-call TTFT and latency in PostgreSQL and Grafana.
- Built self-hosted SIP telephony connecting LiveKit SIP through Asterisk PBX to a Jio trunk, with automated call recording, transcript logging, monitoring, and WhatsApp follow-ups.
- Designed modular debt-collection agents with conversation state management, interruption handling, and silence watchdogs, extensible to customer-service and insurance workflows.

Data Science Intern — **June 2025 – July 2025**
SP Software (P) Limited — *Hyderabad, India*

- Developed automated invoice-extraction pipelines using OCR and deep learning-based layout analysis; fine-tuned TrOCR and Donut for structured fields and trained YOLO region detectors with bounding-box supervision.
- Prepared labeled datasets from internal archives and built post-processing to clean extracted text and validate fields against schema constraints.

Projects

SHL Assessment Recommendation System | *Python, FastAPI, Qdrant* | GitHub | Demo

- Built hybrid retrieval combining dense embeddings and BM25 sparse search with server-side rank fusion in Qdrant and two-stage Cohere/ZeRank reranking to match assessments to hiring queries and job descriptions.
- Added LLM intent parsing for skills, duration, and behavioral/technical requirements, with metadata filtering and selection logic to balance assessment types and penalize duplicates.
- Improved mean **Recall@10 from 0.24 to 0.37** on a labeled query set through retrieval tuning and reranking; deployed a Streamlit UI and FastAPI backend on Render.

Math Routing Agent | *Python, FastAPI, LangGraph, Qdrant, MCP* | GitHub

- Developed a math-tutoring system with RAG over GSM8K and Orca 200k; orchestrated Professor, Critic, Guardrail, and human-feedback stages in LangGraph with conversation memory and bounded iterative refinement.
- Built Qdrant retrieval and MCP web-search tools; compiled a DSPy BootstrapFewShot feedback refiner from collected corrections, with feedback persisted to JSONL and Qdrant.
- Implemented asynchronous human-in-the-loop pause/resume through FastAPI and a React/Vite frontend, allowing critic-rejected answers to be revised from human feedback.

Jev Codebase Explorer (CodeScout) | *Python, Tree-sitter, SQLite FTS5, Git* | GitHub

- Built a local CLI with Git-aware repository discovery and Tree-sitter indexing of Python, JavaScript, and TypeScript symbols; enabled ranked FTS5 search, caller/callee and impact queries, and Git history exploration.
- Implemented an offline heuristic ranker and optional Jev reranking via Vercel AI Gateway; generated deterministic, token-budgeted context packs with exact source spans, related tests, and Git evidence, reporting ranking latency and token usage.

Publication

S. Rishikesh et al., "Leveraging CNN Architectures for Alzheimer's Disease Classification Using Neuroimaging Data," *ICAIIHI*, IEEE, 2025. [IEEE Xplore]

Technical Skills

Languages: Python, Java, SQL, JavaScript
Frameworks/AI: Spring Boot, FastAPI, LangGraph, DSPy, React, PyTorch, Hugging Face Transformers, OCR, RAG
Data/Retrieval: Qdrant, PostgreSQL, MongoDB, SQLite FTS5, BM25, embeddings, reranking
Infrastructure: Kubernetes (K3s), Docker, Git, Redis, Grafana, LiveKit, WebRTC, SIP